Discrete Math W3

Diego Strebel

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ETH Zurich

Hello

I am Diego

3rd Semester CS Student

Material: `https://diegostrebel.com/teaching`

Contact: `distrebel@student.ethz.ch`



Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Where Are We?

⇒ Basic Logic & Proofs

- Sets & Relations
- Number Theory
- Algebra
- Logic

1. Logic & Proofs

- Mathematical Statements
- Concept of a Proof
- Propositional Logic
- Predicate Logic
- Formulas vs. Statements
- Proof Patterns

Universe / Predicate

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A universe is a non-empty set U.

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Definition 2.10

A k-ary **predicate** P on U is a function $U^k \to \{0, 1\}$

Let $U = \mathbb{Q}$

$$\text{prime}(x) = \begin{cases} 1 & \text{if x is prime} \\ 0 & \text{else} \end{cases} \qquad D(x, y) = \begin{cases} 1 & y \neq x \\ 0 & \text{else} \end{cases}$$

$$Q(x, y) = 1 \iff x \text{ and } y \text{ have the same denominator}$$

You can use infix notation

$$x < y : \text{IsLessThan(x, y)}$$

$$\leq \quad \geq \quad \cdot \quad =$$

Quantifiers

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Definition 2.11

For a universe U and a predicate $P(x)$, we define the following logical statements

$$\forall x\, P(x) \quad \text{stands for:} \quad P(x) \text{ is true for } \textbf{all } x \in U$$

$$\exists x\, P(x) \quad \text{stands for:} \quad P(x) \text{ is true for } \textbf{some } x \in U$$

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- $U = \mathbb{N}$: $\forall x\, (x \geq 0)$
- $U = \mathbb{N}$: $\forall x\, \exists y\, (x + y = 0)$
- $U = \mathbb{Q}$: $\forall x\, \exists y\, (x + y = 0)$
- $U = \mathbb{R}$: $\forall x\, (\exists y\, (y^2 = x) \to (x \geq 0))$
- $U = \mathbb{N} \setminus \{0\}$: $\exists x\, \exists y\, \exists z\, (x^2 + y^2 = z^2)$
- $U = \mathbb{N}$: $\exists y\, \forall x\, (x \leq y)$

Interpretation of Formulas

Each formula has parts that are left open for interpretation

Universe, Predicates

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Without fixing an interpretation, a formula is not a mathematical statement!

More in Chapter 6.

Some short exercises

1) Let $U = \mathbb{R}$. What is the meaning of the following formula and is it true or false?

$$\forall x \, (x = 0 \vee \exists y \, (xy = 1))$$

2) What is the meaning of the following formula, and for which universes is it true (or false)?

$$\forall x \, \forall y \, ((x < y) \rightarrow \exists z \, ((x < z) \wedge (z < y)))$$

3) What is the meaning of the following formula, and for which universes is it true (or false)?

$$\exists x \, \forall y \, (x \leq y)$$

Equivalence and Logical Consequence

Same definitions and rules as for propositional logic.

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  $\rightarrow$ *Only holds for identical quantifiers*

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- $\exists x \exists y\, P(x, y) \equiv \exists y \exists x\, P(x, y)$
  $\rightarrow$ *Only holds for identical quantifiers*
- $\forall x\, (x = 0 \lor \exists y\, (xy = 1)) \equiv \forall x\, (\neg(x = 0) \rightarrow \exists y\, (xy = 1))$

Useful Rules for Predicate Logic

- $\forall x\, P(x) \land \forall x\, Q(x) \equiv \forall x\, (P(x) \land Q(x))$

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  - $\rightarrow$ *Note: The converse does not hold:*
    $\exists x\, P(x) \wedge \exists x\, Q(x) \not\models \exists x\, (P(x) \wedge Q(x))$

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Useful Rules for Predicate Logic

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  - $\rightarrow$ *Note: The converse does not hold:*
    $\exists x\, P(x) \land \exists x\, Q(x) \not\models \exists x\, (P(x) \land Q(x))$
- $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
- $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$
- $\exists y \forall x\, P(x, y) \models \forall x \exists y\, P(x, y)$

Useful Rules for Predicate Logic

- $\forall x\, P(x) \land \forall x\, Q(x) \equiv \forall x\, (P(x) \land Q(x))$
- $\exists x\, (P(x) \land Q(x)) \models \exists x\, P(x) \land \exists x\, Q(x)$
  - $\rightarrow$ *Note: The converse does not hold:*
    $\exists x\, P(x) \land \exists x\, Q(x) \not\models \exists x\, (P(x) \land Q(x))$
- $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
- $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$
- $\exists y \forall x\, P(x, y) \models \forall x \exists y\, P(x, y)$
- **BUT:** $\forall x \exists y\, P(x, y) \not\models \exists y \forall x\, P(x, y)$

Logical Formulas vs Mathematical Statements

In general, a logical formula is **NOT** a mathematical statement.

Different interpretations lead to different mathematical statements.

Without fixing an interpretation, a formula is not a mathematical statement!

Examples

Mathematical statements about formulas

- F is valid
- F is unsatisfiable
- $F \models G$

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- F is valid
- F is unsatisfiable
- $F \models G$

$$F \text{ is valid} \implies G \text{ is valid} \qquad (*)$$

$$\not\Updownarrow$$

$$F \models G$$

Examples

Mathematical statements about formulas

- F is valid
- F is unsatisfiable
- $F \models G$

$$F \text{ is valid} \implies G \text{ is valid} \qquad (*)$$
$$\not\Uparrow$$
$$F \models G$$

Lemma 2.4

For any two formulas F and G, if $F \models G$, then $(*)$ is true.

$\models$ vs $\Longrightarrow$

$F \models G$	$S \implies T$
Relation between two **formulas** F and G.	Implication between two **meta-statements** S_1 and S_2.
Example: $F \models G$	Example: F is valid $\implies$ G is valid
Local: G must hold in *every* interpretation where F holds.	**Global**: Assesses whether property S holding guarantees property T.
Cannot be vacuously true unless F is unsatisfiable.	**Vacuously true** whenever statement S is false (e.g., F is not valid).

$\models$ vs. $\Longrightarrow$: Counterexample

To show that (F is valid $\Longrightarrow$ G is valid) does not imply $F \models G$, set

$$F = A \quad \text{and} \quad G = B$$

for atomic variables A and B.

Clearly F is not a tautology and so not valid. Hence the implication is vacuously true.

However one can find an interpretation where $A = 1, B = 0$ and so

$$F \not\models G$$

Proof Patterns

Can be used to prove one step within a longer proof, or sometimes also directly prove a theorem.

Correspond to deduction rules, which are discussed in Chapter 6.

Often many proof patterns will work, **but some make the proof a lot easier.**

Learning when to apply which proof pattern is also a valuable skill to devlop.[1]

[1] For a deeper treatment, I recommend The Book of Proof by Richard Hammack. All following images are taken from this book.

Composition of Implication

Definition 2.12

The proof step of composing implication is as follows: If $S \Longrightarrow T$ and $T \Longrightarrow U$ are true, then $S \Longrightarrow U$ is true

Lemma 2.5

$$(A \to B) \land (B \to C) \models A \to C$$

Direct Proof

Definition 2.13

A direct proof of an implication $S \implies T$ works by assuming S and then proving T under this assumption.

Outline for Direct Proof

Proposition If P, then Q.

Proof. Suppose P.

$\vdots$

Therefore Q. ■

Indirect Proof

Definition 2.13

A indirect proof of an implication $S \implies T$ proceeds by assuming that T is false and proving that S is false, under this assumption.

Lemma 2.6

$$\neg B \to \neg A \models A \to B$$

Outline for Contrapositive Proof

Proposition If P, then Q.

Proof. Suppose $\sim Q$.

$\vdots$

Therefore $\sim P$. ■

Modus Ponens

Definition 2.15

A proof of a statement S by use of the so-called **modus ponens** proceeds in three steps:

1. Find a suitable mathematical statement R.
2. Prove R.
3. Prove $R \implies S$.

Lemma 2.7

$$A \wedge (A \to B) \models B$$

Case Distinction

Definition 2.16

A proof of a statement S by **case distinction** proceeds in three steps:

1. Find a finite list $R_1, \ldots, R_k$ of mathematical statements (the *cases*).
2. Prove that at least one of the R_i is true (at least one case occurs).
3. Prove $R_i \implies S$ for $i = 1, \ldots, k$.

Lemma 2.8

$$(A_1 \vee \cdots \vee A_k) \wedge (A_1 \to B) \wedge \cdots \wedge (A_k \to B) \models B$$

Proof by Contradiction

Definition 2.17

A **proof by contradiction** of a statement S proceeds in three steps:

1. Find a suitable mathematical statement T.
2. Prove that T is false.
3. Assume that S is false and prove (from this assumption) that T is true (a contradiction).

Lemma 2.9

$$(\neg A \to B) \wedge \neg B \models A \quad \text{or equivalently}$$
$$(A \vee B) \wedge \neg B \models A$$

A ”better” way to think about proof by contradiction

Outline for Proof by Contradiction

Proposition P.

Proof. Suppose $\sim P$.

$\vdots$

Therefore $C \wedge \sim C$. ■

A slightly unsettling feature of this method is that we may not know at the beginning of the proof what the statement C is going to be. In doing the scratch work for the proof, you assume that $\sim P$ is true, then deduce new statements until you have deduced some statement C *and* its negation $\sim C$.

If this method seems confusing, look at it this way. In the first line of the proof we suppose $\sim P$ is true, that is, we assume P is *false*. But if P is really true then this contradicts our assumption that P is false. But we haven't yet *proved* P to be true, so the contradiction is not obvious. We use logic and reasoning to transform the non-obvious contradiction $\sim P$ to an obvious contradiction $C \wedge \sim C$.

Existence Proofs

Definition 2.18

Consider a universe $\mathcal{X}$ of parameters and consider for each x in $\mathcal{X}$ a statement, denoted S_x. An **proof** is a proof of the statement that S_x is true for at least one $x \in \mathcal{X}$. An existence proof is **constructive** if it exhibits an a for which S_a is true, and otherwise it is **non-constructive**.

Standard way to prove an $\exists$ statement

Can also be extended to prove uniqueness

WARNING: Although an example suffices to prove an existence statement, a single example does not prove a conditional statement.

Pigeonhole Principle

Theorem 2.10

If a set of n objects is partitioned into $k < n$ sets, then at least one of these sets contains at least $\lceil \frac{n}{k} \rceil$ objects.

We cannot predict or choose which set receives the extra objects. The principle guarantees only existence.



Proof by Counterexample

Definition 2.19

Consider a universe $\mathcal{X}$ of parameters and consider for each x in $\mathcal{X}$ a statement, denoted S_x. A **proof by counterexample** is a proof of the statement that S_x is **not** true for all $x \in \mathcal{X}$, by exhibiting an a (called **counterexample**) such that S_a is false.

Corresponds to existence proof

Used to prove a $\neg\forall P(x) \equiv \exists\neg P(x)$

Note: To disprove an exists statment you have to show $\forall\neg P(x)$

A note on disproving

How to disprove P: Prove $\sim P$.

How to disprove $\forall x \in S, P(x)$.

Produce an example of an $x \in S$ that makes $P(x)$ false.

How to disprove $P(x) \Rightarrow Q(x)$.

Produce an example of an x that makes $P(x)$ true and $Q(x)$ false.

Proof by Induction

Definition

1. *Basis step.* Prove $P(0)$.
2. *Induction step.* Prove that for any arbitrary n we have $P(n) \implies P(n+1)$.

Theorem 2.11

For the universe $\mathbb{N}$ and an arbitrary unary predicate P we have

$$P(0) \land \forall n\,(P(n) \to P(n+1)) \implies \forall n\,P(n).$$

Exercises

Translate Statements to Logic

Let $U = \mathbb{Z}$

Translate the following statements into propositional logic. You may define your own sensible predicates.

1. Every integer has an inverse.
2. The square of every odd integer is odd.
3. For every integer $n > 1$, there exists at least one prime number p such that $n < p < 2n$.
4. There exists exactly one even prime number.
5. There are infinitely many primes p such that $p + 2$ is prime.
6. For every integer $k > 0$, there exist k consecutive composite integers.
7. There are infinitely many positive integers that cannot be written as the sum of three squares.

Describing Formulas

Let $U = \mathbb{Q}$

We define the following predicates

$$P(x) = \begin{cases} 1, & x \in \mathbb{Z} \\ 0, & \text{Else} \end{cases} \qquad Q(x) = \begin{cases} 1, & x < 1 \\ 0, & \text{Else} \end{cases} \qquad R(x, y, z) = \begin{cases} 1, & x + y = z \\ 0, & \text{Else} \end{cases}$$

Describe each of the formulas in words and decide if it is true or false.

1. $\exists x \, (P(x) \land Q(x))$
2. $\exists x \, (Q(x) \land \forall y \, R(x, y, y))$
3. $\forall x \forall y \, (R(x, x, y) \to P(y))$
4. $\forall x \, (Q(x) \to \exists y \, (Q(y) \land R(y, y, x)))$

Direct Proof

Theorem

If n is an odd integer, then $n^2 + 3n + 5$ is an odd integer.

Direct Proof

Theorem

If n is an odd integer, then $n^2 + 3n + 5$ is an odd integer.

Proof.

Since n is odd, we can write $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then we get

$$\begin{aligned} n^2 + 3n + 5 &= (2k+1)^2 + 3(2k+1) + 5 \\ &= 4k^2 + 4k + 1 + 6k + 3 + 5 \\ &= 4k^2 + 10k + 9 \\ &= 2(2k^2 + 5k + 4) + 1 \\ &= 2k' + 1 \quad \text{for some } k' \in \mathbb{Z} \end{aligned}$$

Hence $n^2 + 3n + 5$ is odd. □

Case Distinction

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If $n \in \mathbb{Z}$, then $n^2 + 3n + 4$ is even.

Case Distinction

Theorem

If $n \in \mathbb{Z}$, then $n^2 + 3n + 4$ is even.

Proof.

Assume $n \in \mathbb{Z}$. We proceed by cases.

Case 1: n is even. Then we can write $n = 2k$ for some $k \in \mathbb{Z}$. This gives

$$\begin{aligned} n^2 + 3n + 4 &= (2k)^2 + 3(2k) + 4 \\ &= 4k^2 + 6k + 4 \\ &= 2(2k^2 + 3k + 2) \end{aligned}$$

So $n^2 + 3n + 4$ is even. □

Proof (Continued).

Case 2: n is odd. Then we can write $n = 2l + 1$ for some $l \in \mathbb{Z}$. This gives

$$\begin{aligned} n^2 + 3n + 4 &= (2l+1)^2 + 3(2l+1) + 4 \\ &= 4l^2 + 4l + 1 + 6l + 3 + 4 \\ &= 4l^2 + 10l + 8 \\ &= 2(2l^2 + 5l + 4) \end{aligned}$$

So $n^2 + 3n + 4$ is even.

Since the above cases cover all possibilities, this concludes the proof. □

Indirect Proof

Theorem

Suppose $a \in \mathbb{Z}$. If a^2 is even, then a is even.

Indirect Proof

Theorem

Suppose $a \in \mathbb{Z}$. If a^2 is even, then a is even.

Proof (Indirect).

Assume that a is odd. Then we can write $a = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2k' + 1 \quad \text{for some } k' \in \mathbb{Z} \end{aligned}$$

So a^2 is odd.

By contraposition, if a^2 is even, then a is even. □

Proof by Contradiction

Theorem

$\log_2(9)$ *is irrational*

Proof by Contradiction

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Proof.

Suppose for sake of contradiction that $\log_2(9)$ is rational. Then there exist $a, b \in \mathbb{Z}$ such that $\log_2(9) = \frac{a}{b}$ with $b \neq 0$.

Without loss of generality, we may assume $b > 0$. Since $\log_2(9) > 0$, it follows that $a > 0$. Using logarithm rules, we have

$$\begin{aligned} 9 &= 2^{\frac{a}{b}} \\ 9^b &= \left(2^{\frac{a}{b}}\right)^b \\ (3^2)^b &= 2^a \\ 3^{2b} &= 2^a \end{aligned}$$

Since $a, b \geq 1$, the left side is odd, while the right side is even. This is a contradiction. Therefore, $\log_2(9)$ must be irrational. $\square$

Pigeonhole Principle

Theorem

Any set of seven distinct natural numbers contains a pair of numbers whose sum or difference is a multiple of 10.

Pigeonhole Principle

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Any set of seven distinct natural numbers contains a pair of numbers whose sum or difference is a multiple of 10.

Proof.

Consider the remainders of the numbers when divided by 10. Group the possible remainders $\{0, 1, 2, \ldots, 9\}$ into 6 sets (pigeonholes)

$$\{0\}, \quad \{1, 9\}, \quad \{2, 8\}, \quad \{3, 7\}, \quad \{4, 6\}, \quad \{5\}$$

Assign each of the 7 numbers to the set containing its remainder modulo 10. The Pigeonhole Principle guarantees that at least two numbers, say a and b, belong to the same set.

If a and b have the same remainder, then $a \equiv b \pmod{10}$, so $10 \mid (a - b)$. If a and b have different remainders in the same set, their remainders sum to 10, so $10 \mid (a + b)$.

Thus, there exists a pair whose sum or difference is a multiple of 10. □

Existence Proof: Constructive

Theorem

There exist irrational numbers x, y such that x^y is rational.

Existence Proof: Constructive

Theorem

There exist irrational numbers x, y such that x^y is rational.

Proof.

Let $x = \sqrt{2}$ and $y = \log_2(9)$, both of which are irrational. Then using the properties of logarithms

$$\begin{aligned} x^y &= (\sqrt{2})^{\log_2(9)} \\ &= (\sqrt{2})^{2\log_2(3)} \\ &= \left(\sqrt{2}^2\right)^{\log_2(3)} \\ &= 2^{\log_2(3)} \\ &= 3 \end{aligned}$$

Since 3 is rational, the statement holds. □

Existence Proof: non-constructive

Theorem

There exist irrational numbers x, y such that x^y is rational

Existence Proof: non-constructive

Theorem

There exist irrational numbers x, y such that x^y is rational

Proof.

Let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$. Clearly, y is irrational, but we don't know if x is irrational. We proceed by cases.

Case 1: x is irrational. Then

$$x^y = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^2 = 2$$

So x^y is rational.

Case 2: x is rational. Then $y^y = x$ is rational.

Either way the statement holds. □

Questions?

Bonus: Erdős–Szekeres Theorem

Theorem

Let $n, m \in \mathbb{N}_{>0}$. Then every sequence of $(n-1)(m-1)+1$ distinct real numbers has a monotonically increasing subsequence of length n or a monotonically decreasing subsequence of length m.

Proof of Erdős–Szekeres Theorem (Part 1)

Proof.

Let $(n_1, n_2, \ldots, n_N)$ be a sequence of $N = (n-1)(m-1)+1$ distinct real numbers. Label each term n_i with an ordered pair (a_i, b_i), where a_i is the length of the longest monotonically increasing subsequence ending at n_i, and b_i is the length of the longest monotonically decreasing subsequence ending at n_i.

To show that every term receives a distinct label pair, consider any two indices $i < j$:

If $n_i < n_j$, any increasing sequence ending at n_i can be extended by n_j, so $a_i < a_j$.

If $n_i > n_j$, any decreasing sequence ending at n_i can be extended by n_j, so $b_i < b_j$.

Thus, $(a_i, b_i) \neq (a_j, b_j)$ whenever $i \neq j$, giving N distinct pairs of labels. □

Proof of Erdős–Szekeres Theorem (Part 2)

Proof (Continued).

Suppose for sake of contradiction that there is no increasing subsequence of length n and no decreasing subsequence of length m. So we have $a_i \leq n-1$ and $b_i \leq m-1$. Since $a_i, b_i \geq 1$, there are only $(n-1)(m-1)$ possible label pairs.

By the Pigeonhole Principle, placing $N = (n-1)(m-1)+1$ distinct label pairs into $(n-1)(m-1)$ possible pairs is impossible. Therefore, $\exists k \in \mathbb{N} \setminus \{0\}$ such that $a_k \geq n$ or $b_k \geq m$.

If $a_k \geq n$, then by definition, the longest increasing subsequence ending at n_k has length at least n, which guarantees an increasing subsequence of length n. Similarly, if $b_k \geq m$, the longest decreasing subsequence ending at n_k has length at least m, which guarantees a decreasing subsequence of length m.[2] □

[2] See: Seidenberg, A. (1959), "A simple proof of a theorem of Erdős and Szekeres"